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 SURVEY

# LLMs Integration in Recommender Systems: A Comprehensive Survey of Frameworks, Taxonomies, and Applications

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**ABSTRACT** Recently, Large Language Models (LLMs) have proven to be powerful tools for tackling persistent challenges within recommendation systems. This systematic literature review (SLR) explores how LLMs are being integrated into recommender systems. By examining studies published between 2018 to 2025, this study aims to provide a comprehensive overview of existing approaches, real-world applications, and recent advances in this area. The study addresses three research questions including existing methodologies and frameworks for integrating LLMs into recommender systems, developing a comprehensive taxonomy of LLM-enhanced recommender systems, and examining domain-specific applications across a range of business contexts. Through our SLR method, we examined 55 primary studies and identified five major integration approaches: CLLM4Rec-based frameworks, knowledge graph-driven integration, LLMER-type architectures, the strategic placement of LLMs at various stages of the recommendation pipeline, and purpose-built frameworks such as LLM4Rec. To address the second research question, we introduce a taxonomy that captures cold-start mitigation approaches, integration strategies including embedding-based, prompt-based, fine-tuned, and agent-oriented methods, as well as system architectures and evaluation criteria. The third research question examines the practical impact of these systems across a range of real-world application domain. Recent studies indicate that recommendation systems powered by LLMs are reshaping sectors such as e-commerce, retail, news media, content platforms, and healthcare. In these settings, LLMs contribute to more accurate product recommendations, more personalized marketing initiatives, and stronger customer engagement. In healthcare contexts, these models assist practitioners by supporting patient data analysis and informing clinical decision-making processes. Our results indicate that recommender systems enhanced with LLMs achieve strong performance, while demonstrating an improved ability to capture semantic information and nuanced contextual relationships. This comprehensive review provides researchers and practitioners with structured insights into recent advances in recommender systems powered by LLMs.

**INDEX TERMS** Large language models (LLMs), cold-start problem, recommender systems, systematic literature review (SLR), taxonomy, framework.

## I. INTRODUCTION

Traditionally, human's ability to access knowledge and information has been heavily restricted by both technological

barriers and practical constraints. There was always a clear gap between how much information people wanted and what was actually available, since knowledge sources like books, academic papers, and reports were scarce and often expensive. This situation changed dramatically when we entered the digital era. The massive expansion of digital

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content and the proliferation of online platforms completely reversed the old problem of scarcity, leading instead to what we now recognize as a new challenge: Information Overload [1].

In recent years, information overload has been increasingly recognized as a major challenge in modern information systems [2]. Individuals are continuously exposed to large volumes of information from multiple sources, often regardless of their intent to seek it. This rapid expansion generates substantial noise at critical levels. Major platforms such as YouTube, Amazon, Spotify, eBay, and Netflix are particularly impacted, as they continuously collect user interaction data including clicks, views, reads, transactions, and other forms of implicit feedback [3]. This data-driven approach to modeling user preferences [4] has proven effective in the development of personalized recommender systems that deliver meaningful and relevant recommendations [4].

One of the ongoing challenges in practical recommender systems is generating useful recommendations when there's no previous user interaction data available—a problem researchers call the “cold-start” issue [4]. This issue typically emerges in multiple context including recommendation for new users, recommend brand-new items, as well as recommendations for deploying systems with sparse historical records [5]. Despite significant research, the cold-start problem continues to be a key technical and practical challenge, particularly in new and rapidly evolving application domains [6], [7], and hindering the effectiveness and widespread use of recommender systems, especially on new platforms and in fast changing areas [8].

To improve the performance and robustness of recommender systems, a range of advanced computational algorithms and techniques have been developed over time. To enhance precision of recommendation systems collaborative filtering methods, deep learning techniques, and graph neural network have been investigated [9]. However, large language models (LLMs) have recently become a major focus of attention in both academic research and industrial applications [5]. LLMs represent a novel paradigm in artificial intelligence, demonstrating exceptional proficiency across various language processing tasks and considerable promise to enhance recommender systems.

LLMs exhibit unique features that make them effective solutions for enduring issues in recommendation and data mining. These capabilities arise from LLMs' extensive collections of open-world knowledge encoded during pre-training, advanced logical and commonsense reasoning, and a profound understanding of human society, culture, and behavioral patterns [6]. Moreover, sophisticated methodologies such as in-context learning and prompt engineering enable LLMs to achieve robust performance on downstream tasks without requiring task-specific fine-tuning [5]. Thus, it is increasingly advantageous to develop next-generation recommender systems based on pretrained LLMs, utilizing

their knowledge repositories, reasoning abilities, and generative AI capabilities [10].

Recent studies indicate that LLMs can be successfully utilized in several recommendation contexts. LLMs have demonstrated potential as zero-shot rankers, i.e., ranking candidate items without task-specific training [11]. They perform effectively as conversational agents for interactive, clarifying recommendations, facilitating more engaging and interpretable user experiences [12]. LLMs have demonstrated effectiveness in mitigating the cold-start problem by simulating user preferences and employing data augmentation methodologies [13]. Furthermore, LLMs may improve data procedures, thereby enhancing the efficacy of recommender systems when substantial amounts of textual data are available [14], [15].

When significant amounts of textual information are accessible, LLMs can integrate this varied textual data to offer more accurate, contextually informed, and interpretable recommendations, particularly within content-rich domains characterized by detailed descriptions and user-generated comments [9], [16]. The integration of LLMs with knowledge graphs has enhanced their recommendation capabilities, facilitating more organized, semantically informed methodologies [17]. Moreover, agentic recommender systems utilizing multimodal LLMs represent an emerging domain, integrating language understanding with recommender algorithms to develop more independent and adaptable systems [18].

The integration of LLMs and recommender systems (RSs) is an evolving, active research domain with significant practical implications. This integration offers new opportunities to improve recommendation effectiveness and enhance the user experience [19]. Additionally, it delivers more interpretable and explainable recommendations, addresses enduring challenges such as the cold-start problem, and enables personalized recommendations across diverse application domains [20], [21]. While LLMs keep improving and research examines both discriminative and generative paradigms [10], their use in recommender systems could change how recommendations are made across diverse industries and domains.

Recent surveys and studies have examined how LLMs are being integrated into recommender systems [19], [22], [23]. They highlight both the exciting potential of these models and their challenges such as hallucination issues [24], concerns about robustness [25], and the ongoing need for explainability [26]. While these reviews provide helpful overviews of the field, there is still a clear need for deeper, more systematic work, particularly in organizing methodological approaches, developing structured taxonomies, and comprehensively mapping domain-specific applications.

This systematic literature review (SLR) aims to: (1) Examine the rapidly evolving relationship between LLMs and recommender systems within the current research landscape; (2) Identify, categorize, and analyze the methodologies,

frameworks, and strategies used to integrate LLMs into recommendation pipelines; (3) Develop a structured and comprehensive taxonomy of LLM-enhanced recommender systems based on their methodological approaches and system architectures; and (4) Explore the practical domains in which LLMs improve recommendation performance, including e-commerce, entertainment, education, healthcare, cultural heritage, and emerging sectors. The review provides an evidence-based understanding of current developments and highlights promising directions for future research and innovation in this rapidly advancing field.

The paper proceeds as follows. Section II introduces and reviews recommender systems, LLMs, and their integration. Section III outlines utilized research methods in this study. Three underlying research questions are investigated in three subsequent sections. Conclusion is presented in Section VII. Subsequently, emerging subfields and future work are provided in Section VIII.

## II. RECOMMENDER SYSTEMS, LARGE LANGUAGE MODELS, AND THEIR INTEGRATION

The following three subsections provide background information on Recommender Systems, LLMs, their integration, and real world implementation considerations.

### A. RECOMMENDER SYSTEMS

Recommender systems are algorithms that search through large volumes of data for certain contents, products, or services that may be appealing to a particular user due to their past behavior, preference, and trends [27]. They have played an important role in addressing the challenge of information overload in the modern digital era [1]. Reference [3] provides a through overview of classical methods and a discussion on the evolution of recommender systems. Classical recommender systems have been developed based on three foundational paradigms, namely, Collaborative filtering, Content-based methods, and Hybrid approaches.

Collaborative filtering, discussed in [28], is based on the user-item interaction patterns and assumes that users with similar preferences will consider similar items. The collaborative filtering paradigm encompasses both memory-based and model-based methods.

Content-based methods recommend items based on their attributes and past preferences that the user has for similar items. Content-based systems excel in cases where user-item interaction data is sparse and fail to recommend new items quite different from what the users have already consumed.

Hybrid recommenders combine several techniques to take advantage of complementary strengths. They mostly outperform by overcoming the individual approaches through ensemble learning and weighted combination strategies. Examples include collaborative filtering and deep learning-based [29], knowledge-based recommendation [30], network architectures and graph-based approaches [9], variational bandwidth auto-encoders [31], neural collaborative filtering and several encoder-decoder architectures [32].

Recent studies reveal that there is a paradigmatic shift towards more dynamic and interactive mechanisms of recommendation. Reference [33] introduces generative recommendation paradigms and suggests that recommender systems should move from ranking-based approaches to generative models that can provide adaptive personalized recommendations. Reference [34] proposes a unified multi-task learning framework for conversational recommender systems with multiple goals. One significant development direction is conversational recommendation. Reference [35] proposes a unified conversational recommender system using knowledge-enhanced prompt learning, illustrating how dialogue-based interfaces allow nuanced preference discovery. Reference [36] developed RecInDial, a unified framework that enables conversational recommendation with pre-trained language models; this bridges traditional recommendation techniques with natural language interfaces. Reference [37] discusses topic-guided conversations in recommender systems, incorporating topical structure into the dialogue-based recommendation mechanisms.

It should be noted that, even with all their enhancement, modern recommender systems deal with several interconnected challenges including -Data sparsity where many items have limited interactions, -Diversity and fairness, especially in limiting the visibility for emergent content [38], -Temporal dynamics as user preferences evolve over time and item relevance changes contextually, and -Explainability and transparency [39].

Cold-start is considered as another critical and challenging problem in recommender systems. Reference [4] explained that cold-start issues occur in three settings: new user, new item, and when both users and items are new. User cold-start arises when a new user joins a system, usually without interaction history. The traditional collaborative filtering approaches fail since they require large user-item interaction data to compute similarities. New item cold-start presents challenges when newly added items do not have enough interaction data for recommendation [40]. The phenomenon becomes particularly problematic in areas with high item turnover, such as e-commerce and news platforms, where new products or information are constantly included into the catalog [41]. [42] conducted a systematic review of the user cold-start problems and categorized the state of the art approaches into content-based methods, knowledge-based methods, and hybrid approaches. Recent innovations promise solving cold-start issues. Reference [43] proposed task-adaptive neural processes for user cold-start recommendation using meta-learning approaches. However, recent advancements in deep learning led to interesting solutions, such as DropoutNet, which employs content-based features and neural architectures to integrate content and collaborative information, providing successful recommendations regardless of insufficient or no interaction data [44]. Despite these advancements, the cold-start problem remains a major research challenge. The quality of recommendations for new or unseen items plays a critical role in shaping

user satisfaction and key business outcomes in real-world applications.

## B. LARGE LANGUAGE MODELS

LLMs represent a new paradigm in natural language processing and artificial intelligence. Reference [6] gives a broad overview survey of LLMs, defining them as neural network models that have been pre-trained on enormous volumes of text that can understand and generate human language with exceptional fluency and contextual awareness. These models have mainly utilized transformer architectures [45], replacing recurrent neural networks with attention-based mechanisms. They have demonstrated outstanding performance for various tasks related to natural language understanding and generation tasks. Reference [46] surveys transformers and ensemble learning approaches in natural language processing (NLP). This indicates the impact of transformer-based architectures on the performance of NLP tasks. With increasing model size, larger models have better performance on different tasks, particularly in language reasoning, though much more expensive in computation.

According to [47], LLM architectural paradigms are generally in three forms, namely, encoder-only, decoder-only, and encoder-decoder. Encoder-only models excel at user/item representation learning. Decoder-only models drive conversational and generative recommendation. Encoder-decoder architectures support sequence-to-sequence tasks such as review summarization and explanation generation. Reference [48] performed a survey on evolution, architectures, adaptation, benchmarking, applications, and challenges of LLMs.

[49] stated that LLMs implicitly encode a significant amount of factual knowledge within their parameters through pre-training on text corpora. However, this implicit knowledge creates challenges in extraction and verification. Reference [17] proposed a roadmap to unify LLMs and knowledge graphs. The reasoning capability of LLMs still remains an active research area. For instance, [50] introduced Graph-of-Thought, which utilizes LLMs to solve complex and dynamic business problems based on structured reasoning processes.

Pre-trained LLMs have strong zero-shot abilities. However, fine-tuning on task-specific data can improve performance in specialized areas. Reference [51] presents a prompt tuning method that adjusts only a few learnable parameters while keeping the main model unchanged. This reduces the required computation and allows for efficient adaptation to multiple tasks without needing different model copies. Another effective approach is prompt-based adaptation, or so-called in-context learning, [52]. Additionally, domain-specific fine-tuning on specialized corpora can improve performance on specialized domains, e.g., [53]. More recently, development of LLMs has gone beyond text, such as image captioning, visual question answering, and document understanding, to include multimodal information;

for instance, [54] infuses behavioral science into the LLM for activity coaching.

Despite remarkable capabilities, LLMs face significant limitations that hinder their deployment. A few of those are summarized below.

*Hallucination:* One limitation of LLMs is that they often generate plausible factually incorrect information, which is referred to as a kind of hallucination. A survey of hallucination in natural language generation is presented in [24]. The study in [55] assesses ChatGPT along the dimensions of reasoning, hallucination, and interactivity. The findings are that state-of-the-art models have significant hallucination rates for factual reasoning tasks. Recently a survey on taxonomy, causes, detection and mitigation strategies of hallucination is provided by [56].

*Robustness:* LLMs are still questioned regarding their robustness in adversarial and out-of-distribution scenarios. The research by [25] investigates how robust ChatGPT is when faced with adversarial inputs, its weaknesses to such inputs, changes in distribution, and crafted prompts that may induce undesirable behaviors.

*Explainability and Interpretability:* One of the significant issues needing further research is LLMs explainability, especially when they are applied in high-stakes problems. The survey by [39] focused on explainable AI in NLP. It stated that model interpretability is importance in applications involving individual decision-making or safety-critical domains. Also, [26] comprehensively surveyed explainability for LLMs and some frameworks explaining the decision-making process of such a complex model.

*Creativity and Serendipity:* LLMs' capacity for combinational and exploratory creativity can enhance serendipitous recommendations, helping users discover items outside their typical preference patterns. However, as noted by [57], LLMs exhibit limitations in transformational creativity due to their autoregressive, distribution-following nature. In other words, while LLMs can generate creative explanations and suggest unexpected item combinations, they may still be bounded by patterns in their training data.

*Ethics and Transparency:* Ethical issues, such as bias propagation from training data, the potential for generating deceptive content, and environmental costs due to large scale training and deploying, remain major concerns. Reference [58] surveys legal aspects of LLMs including questions about copyright, liability, and regulatory frameworks as LLMs become widespread. A serious challenge to deployment in regulated domains arises from the lack of interpretability of neural networks, requiring algorithmic decision-making transparent justification.

## C. LLMs IN ENHANCING RECOMMENDER SYSTEMS

The integration of LLMs into recommender systems represents a paradigm shift in how suggestions are generated, explained, and personalized. A comprehensive survey by [22] discusses various ways recommender systems can leverage



LLMs. By integrating LLMs, many traditional limitations of recommender systems are circumvented and new capabilities have been added. Several recent surveys have provided systematic overviews of this growing field: [21] discusses the impact of LLMs on recommender systems in depth, [23] conducts a comparative analysis of methodologies, [20] provides extensive coverage of LLM-powered systems in multiple paradigms, and [8] surveys the approaches, applications, and challenges in the LLM4Rec domain. Additional views on how LLMs enhance recommendation effectiveness are given in [7] and [59].

The following subsections outline instances that LLMs are integrated in recommender systems. Through these applications and approaches, LLMs have significantly improved the effectiveness of recommender systems, mitigating most of the long-standing challenges and introducing new opportunities for personalization, explainability, and interactivity. The integration of LLMs represents not only an incremental improvement but also a fundamental reconceptualization of how recommender systems work and interact with users.

### 1) ZERO-SHOT AND FEW-SHOT RECOMMENDATION

The major contribution of LLMs to recommendation is the capability for recommending with no task-specific training data. Recent study of [11] have demonstrated that LLMs may work as zero-shot rankers of recommender systems, whose ranking of items relies merely on understanding the language without having recommendation-specific training data. This stems from their ability to understand natural language descriptions of items and users, employing their implicit preference understanding in ranking tasks. Reference [60] explores LLMs as zero-shot conversational recommenders, where they can make recommendations through natural language dialogues with minimal training on recommendation-specific data. Furthermore, [12] introduce M6-Rec, which demonstrates that generatively pre-trained language models can serve as open-ended recommender systems capable of generating diverse recommendations in different domains. Between the extremes of zero-shot and fully supervised learning, few-shot recommendation has emerged as a promising intermediate approach, where models learn to make a recommendations from just a few examples [61]

### 2) ADDRESSING COLD-START PROBLEMS

[4] stated cold-start as a fundamental limitation when new users or items do not have sufficient historical data. Its severity increases with the proportion of new users in the system and the rate of new item introduction. Recent work leverages LLMs to fundamentally address this challenge through multiple complementary mechanisms.

[62] explores whether small language models can function as recommenders for data-centric cold-start recommendation. It shows that even smaller variants of LLMs can make meaningful recommendations with extremely minimal data;

this democratizes the use of LLM-based recommendation to resource-constrained environments. Reference [16] proposes LLMTreeRec, which leverages the powers of LLMs for cold-start recommendations by knowledge-enhanced tree structures. Organizing items in hierarchical relations, the system utilizes the LLM capacity of understanding categorical relationships and browses the item spaces for recommendations that are completely new.

[63] uses LLMs to interpret user interests from limited information, solving cold-start item recommendation problems. The approach leverages language understanding to extract meaningful preference signals from minimal information provided by users and reduce data requirements for effective recommendation. Reference [13] presents a novel LLM simulator approach for cold-start recommendation. In this work, LLMs generate synthetic user behaviors and preferences, which are employed to bootstrap recommender systems. This data augmentation strategy mitigates data scarcity by creating synthetic interaction data reflecting realistic user behavior patterns.

A novel framework is proposed in [64], specifically designed for cold-start advertising systems powered by LLMs, extending cold-start solutions to the advertising domain where novelty and relevance are particularly challenging. Reference [65] demonstrates how LLMs act as data augmenters for cold-start item recommendation by generating synthetic training data that bypass the data scarcity. The generated data can be annotated with pseudo-ratings, offering training signal for classic traditional collaborative filtering methods without requiring any real user interactions.

### 3) CONVERSATIONAL AND INTERACTIVE RECOMMENDATIONS

LLMs enable natural conversational interfaces for recommender systems, transforming user experiences from passively consuming ranked lists to having an active dialogue about preferences. Chat-Rec is an interactive and explainable recommender system proposed by [5]. It enhanced user-system dialogues where recommendations emerge through conversation.

A comprehensive survey on conversational recommender systems is conducted by [66]. It highlights the critical role played by interactive preference elicitation through dialogue. Conversational methods allow users to tune their preferences iteratively, ask for clarification, and offer personalized suggestions based on dialogue context. Reference [67] presents a conversational recommender system enhanced with LLM, maintaining context through long dialogues. It allows the system to track user preference evolution over time, remember constraints mentioned previously, and recommend items smoothly as dialogue progresses. The recommender AI agent was proposed in [68] that combined LLMs with such interactive recommendations utilizing user feedback. Agents can now execute multi-turn recommendation processes autonomously, from information gathering

to preference clarification, item search, and explanation generation—all guided by user feedback.

The advantages of the conversational paradigm include: -reducing cognitive load by eliminating the need to scan long recommendation lists, -enabling the specification of complex preferences which could be hard to indicate in structured forms, and -preference discovery, whereby users learn about options through the dialogue. The interaction also develops trust because of transparency and the personalization involved.

#### 4) EXPLAINABILITY AND REASONING

A significant benefit of incorporating LLMs within recommender systems is much superior explainability. The work by [5] emphasizes explainable aspects of LLM-augmented systems whereby natural language explanations of recommendations are given to users in transparent human language instead of incomprehensible algorithmic scores.

Recommendation is framed by [69] as instructions, where LLMs literally understand and follow user preferences, either explicitly stated or implicitly inferred from the dialogue context. This framing stresses that recommendations arise from comprehension and execution of user intent, entailing transparency within the mapping between preferences and suggestions. [70] applies LLMs and step-back prompting to create robust legal case recommender systems with improved interpretability. The step-back prompting technique decomposes complex reasoning into interpretable steps that enables users to trace the reasoning chain from case characteristics to recommendations.

Explainability enables a set of downstream benefits in recommendation systems. Users can check recommendation appropriateness and make corrections when explanations reveal misunderstandings of their preferences or intent [27], [28], [66]. For system engineers, explainability helps identify and reduce biases, false correlations, or undesired patterns in recommendation algorithms. This, in turn, supports debugging and model improvement [26], [39]. From a governance standpoint, clear explanations help regulators and auditors to assess compliance with legal and ethical standards, including fairness and accountability in algorithms [26], [58]. Moreover, explainable recommendations boost users' trust, confidence, and sense of control while using intelligent systems. This is particularly important in complicated recommender systems that use LLMs [5], [21], [28].

#### 5) PROMPT ENGINEERING AND PERSONALIZATION

Prompt engineering is the process of crafting clear instructions to guide generative AI models toward producing desired outputs. Prompt engineering involves various methods: few-shot prompting, which involves giving examples of desired recommendation behavior; chain-of-thought prompting, which involves encouraging step-by-step reasoning about recommendation decisions; and role-based prompting,

which establishes system personas that encourage certain recommendation styles. Well-thought-out prompt design has a significant impact on recommendation quality without requiring model retraining.

Effectively integrating LLMs into a recommendation setup, in fact, requires sophisticated approaches to prompt engineering that guide model behavior toward application-specific objectives. This study [71] comprehensively explores prompt engineering for LLMs and elaborates on how to elicit optimal performance from them. Therefore, an effective prompt would need to communicate the task at hand, the desired format of the output, preference constraints, and any contextual information.

This study [72] proposes reinforced prompt personalization for recommendations with LLMs, with prompts dynamically adapted to individual user characteristics. Personalized prompts substantially improve recommendations, and they are different from generic ones since they include user-specific information, previously expressed preferences, and interaction history.

#### 6) DATA AUGMENTATION, RANKING, AND RETRIEVAL

The contributions of LLMs to recommender systems occur through multiple complementary mechanisms beyond suggestion generation. Data augmentation approaches create synthetic training data to address data scarcity, enabling traditional collaborative filtering methods to operate effectively despite only limited real interactions.

RecRanker, proposed by [73], uses instruction-tuned LLMs specifically in ranking top-k recommendations. Instead of generating recommendations directly, the system reranks candidate items using LLM understanding of user preferences and item properties, combining candidate generation efficiency with ranking precision. Reference [74] discusses retrieval-augmented LLMs for enhanced recommender systems by including LLM capabilities with external knowledge retrieval. Retrieved information anchors recommendations in verifiable knowledge, reducing from hallucination, and improving reliability. DALLRec [15] is an effective data augmentation framework in fine-tuning LLMs for recommendation. DALLRec synthesizes user-item interaction data based on a given sparse real interaction dataset for better model training.

Ranking and re-ranking approaches leverage the strengths of LLMs without requiring a full replacement of existing recommendation pipelines. This modular nature of the integration allows for more gradual adoption and is more easily backward-compatible with operational systems. Similarly, retrieval-augmented methods also maintain existing item generation mechanisms and just refine the ranking using LLM-based filtering.

#### 7) MULTIMODAL AND AGENTIC SYSTEMS

Recent works extend LLM-based recommendation to multimodal contexts involving multiple diverse data types. Agentic

recommender systems in multimodal LLMs are addressed by [18]. These systems can process images, text, audio, and structured data simultaneously, and reason over them. Multimodal understanding allows for richer item descriptions and user profiles for better recommendation precision.

Large Language and Graph Assistant (LLaGA), introduced in [75], incorporates LLMs with graph-structured information to enhance recommendations. Graph structures can model user-item relationships, social networks, and hierarchical item categories. Combining the language understanding of LLMs with the structured reasoning of graph neural networks creates complementary capabilities.

[76] presents collaborative LLMs into recommender systems, allowing various models to work cooperatively. Ensemble approaches that combine several LLMs, each specialized for particular aspects, may yield better recommendation results due to through diversity and specialization.

Agentic approaches allow LLMs to independently execute complex recommendation workflows. Instead of the single-pass generation of recommendations, agents can iteratively gather information, evaluate candidate items, generate explanations, and refine recommendations based on system feedback and user reactions.

#### 8) DOMAIN-AGNOSTIC AND TRANSFER LEARNING APPROACHES

A newly emerging paradigm is that LLMs can work as universal recommendation engines applicable across diverse domains. As pointed out by [77], LLMs are domain-agnostic recommender systems that adapt to various recommendation domains without substantial retraining. Recent studies further support this feature, showing that LLMs are capable of effective cross-domain knowledge transfer, allowing them to bridge the semantic gaps that often exist across different recommendation contexts [78]. Building on this foundation, [79] present the Universal Recommendation Model (URM), which integrates multiple previously isolated recommendation tasks into a cohesive input-output framework utilizing task prompts. This approach facilitates zero-shot task transfer and allows for efficient rapid modification in large-scale industrial systems. The concept of universality stems from the fact that LLMs are pre-trained on diverse data of various domains such that transfer learning across domains is possible without explicit domain-specific training. In contrast to conventional recommendation models that depend on ID-based representations, LLMs utilize semantic comprehension of item descriptions and user preferences, enabling them to align conceptually analogous items across diverse domains and promote efficient information transfer [77], [79].

The domain-agnostic perspective has considerable practical benefits: single models can support multiple domains of recommendation tasks simultaneously, development resources are focused on prompt engineering and light fine-tuning rather than domain-specific model development, and

knowledge transfer in newer domains benefits from extensive developments in mature ones. Furthermore, these universal approaches reveal particular advantages in addressing data sparsity and cold-start issues by using cross-domain behavioral patterns and semantic correlations [78].

#### 9) SPECIALIZED APPLICATIONS AND DOMAIN EXTENSIONS

LLMs have been applied successfully to specialized recommendation domains that require domain expertise and nuanced understanding. For example, [80] used LLMs as recommender systems in museums for personalized recommendations on cultural heritage. The work in [81] describes crafting museum guides using ChatGPT4, showing practical deployment of LLM-based recommendations for cultural institutions.

[45] applies recommender systems based on LLMs in higher education, specifically for literature recommendations in student term paper supervision. The domain application shows that LLMs understand academic topics, assess the relevance of the papers to students' needs, and give pedagogically appropriate suggestions.

[82] leverages generative LLMs in recommender systems for dynamic intention understanding through case study analysis of interactions between clients and designers. It shows how such models are able to understand implicit user intentions from conversational context and utilize that understanding for recommendation improvement.

[83] discusses how fine-tuned LLMs can be applied to reduce data requirements and enhance data efficiency in analytical tasks such as load profile analysis in energy systems. This application further develops the usage of LLMs beyond traditional recommendation towards specialized data-constrained analytics tasks.

#### 10) INTEGRATION WITH KNOWLEDGE STRUCTURES AND REASONING

Merging LLMs with structured knowledge provides a strong direction for addressing hallucination concerns and improving reliability. Reference [17] proposes a holistic roadmap that unifies LLMs and knowledge graphs, showing how structured knowledge can enhance LLM-based reasoning and ground outputs in verifiable facts. Ground is the process of connecting a model to specific, reliable data sources to improve the accuracy and relevance of its responses.

Graph-of-Thought is introduced in [50], an architecture that leverages LLMs in solving complicated, dynamic business problems by structured reasoning processes. Explicit structuring of the reasoning as graph traversal allows these systems to break complex problems into tractable subproblems and yield better solutions. Similarly, for industrial recommender, [84] presents an approach using LLMs with inferential knowledge graphs that enhance reliability and traceability in enterprise settings. Knowledge graphs represent structured relationships among items and attributes by utilizing LLM recommendations.

### 11) HUMAN FEEDBACK AND ITERATIVE IMPROVEMENT

Recent advances illustrate that human feedback mechanisms can enhance LLM-based recommender systems through continuous learning and adaptation. Reference [85] propose learning from human feedback from LLMs for content quality-aware recommendation is discussed in. Instead of relying solely on automated metrics, systems can learn human preferences through active learning and preference elicitation. These approaches constitute a foundation towards more human-centered AI, where the systems improve utilizing users feedback. Such an iterative improvement cycle enables recommender systems to adapt to user preferences and fine-tune their understanding for a diverse populations of users. This approach indicates a significant shift toward more human-centered AI systems that evolve through frequent user feedback loops. Building on this foundation, [86] introduce Lusifer, an LLM-based simulation environment that creates realistic, dynamic user input for training recommender systems based on reinforcement learning. In this framework, user profiles are updated step by step with each encounter. LLMs give clear explanations of how and why preferences change over time, which helps systems better understand how people's preferences change. Moreover, [87] highlight the significance of refined negative feedback by employing LLMs to differentiate between falsely labeled negative feedback (which includes positive indicators) and confirmed negative feedback, thus enhancing the quality of training signals and the accuracy of recommendations. This human-in-the-loop methodology not only enhances the quality of recommendations but also fosters user trust through transparent feedback mechanisms and adaptive learning.

### D. IMPLEMENTATION CONSIDERATIONS

Despite practical performance reported across the studies reviewed in this SLR, commercializing research prototypes creates distinct opportunities as well as a set of challenges. A few instances are provided below.

*Architectural Integration:* Majority of popular recommender pipelines are established based on collaborative filtering or matrix factorization infrastructures. Thus, replacing these with LLM-based systems is neither practical nor necessary. Instead, modular integration strategies using LLMs as re-rankers, conversational front ends, or feature augmentation layers, are suggested. This allows organizations to utilize existing investments while enhancing specific stages, [22], [73], [74]. Also, scalability concerns can be considerably mitigated through parameter-efficient fine-tuning techniques such as Low-Rank-Adaptation (LoRA) and prompt tuning [51].

*Latency:* It is still one of consequential feasibility burdens. Including LLMs in recommender system can cause higher latency, making synchronous real-time recommendations challenging [7]. Some attempts such as Retrieval Augmented Generation architectures partially address this concern by separating the knowledge retrieval stage from generative

inference [74]. Complementary strategies such as response caching, asynchronous inference pipelines, and model distillation can also reduce per-query latency [15], [77].

*Cost:* On the cost factor, fine-tuning a pre-trained open-source LLM on domain-specific interaction data is even cheaper than training a specialized model from scratch [51], [71]. Industry-level deployments reviewed in this SLR, such as e-commerce platforms, healthcare decision support systems, and education recommendation tools, collectively demonstrate that these mitigation strategies are adequate for production deployment [45], [53], [59]. As a result, since domain-adaptable LLMs continue to mature, real-world feasibility is expected to improve further. This makes LLM-enhanced recommender systems a practical choice for a broad range of organizational contexts.

*Computational Efficiency Strategies:* Researchers have explored several techniques to make LLM-based recommender systems more practical and less computationally expensive. One widely used approach is model distillation, where a large model is compressed into a smaller and faster version with minimal loss in performance [15], [77]. Quantization is another technique that reduces the precision of model weights, making deployment more efficient, especially on resource-constrained hardware.

To lower training costs, parameter-efficient fine-tuning methods such as LoRA and prompt tuning update only a small subset of the model's parameters instead of retraining the entire model [51]. This significantly reduces computational overhead. In addition, mixture-of-experts architectures improve efficiency by activating only a portion of the model for each query, rather than using all parameters at once.

Beyond model-level optimizations, system-level strategies also play an important role. Techniques such as response caching, asynchronous inference pipelines, and batch processing help reduce latency and improve system throughput [74].

*Privacy, Security, and Regulatory Considerations:* LLM-enhanced recommender systems depend heavily on personal user data, such as interaction histories, preferences, and behavioral patterns. While this enables more accurate personalization, it also introduces important privacy and security concerns. To address these issues, researchers have explored privacy-preserving techniques like federated learning, where models are trained locally on user devices without sharing raw data, and differential privacy, which adds controlled noise to protect individual identities [58]. In addition, approaches such as data anonymization and secure aggregation can help safeguard sensitive information during both training and inference.

From a regulatory standpoint, compliance with frameworks such as the General Data Protection Regulation and the California Consumer Privacy Act is critical. These regulations emphasize requirements like obtaining user consent, minimizing data collection, and ensuring the right to erasure. At the same time, the lack of interpretability of LLMs presents challenges for transparency, as regulations



increasingly demand explainable and accountable decision-making processes [26]. Overall, future systems must carefully balance the benefits of personalization with strong privacy protections in order to maintain user trust and meet evolving regulatory requirements.

*Scalability in Large-Scale Deployments:* Scaling LLM-enhanced recommender systems for real-world use introduces challenges related to inference throughput, memory usage, and latency [7]. To address these issues, several practical strategies have been proposed. For instance, modular architectures can reduce computational overhead by restricting the use of LLMs to specific stages of the pipeline, such as re-ranking [22], [73].

In addition, universal models can handle multiple tasks through prompt-based inputs, improving both flexibility and efficiency [79]. Parameter-efficient fine-tuning methods further reduce training and deployment costs by updating only a small subset of model parameters [51], [77]. Another effective approach is retrieval-augmented generation, which separates retrieval and generation components, allowing them to scale independently [74].

Findings from industry deployments included in this systematic literature review suggest that these strategies are already practical for production environments [45], [53]. However, further research is still needed to improve performance under high-concurrency and real-time conditions.

*Sustainability and Energy Efficiency:* The high energy demands of training and deploying LLMs have raised growing environmental concerns. To address this, several strategies can help improve efficiency while reducing the overall carbon footprint. For example, techniques such as model distillation, quantization, parameter-efficient fine-tuning [51], and mixture-of-experts architectures not only lower computational costs but also make these systems more energy efficient.

In addition, using smaller, task-specific models when appropriate can further reduce resource consumption without significantly compromising performance [61]. Improvements in inference infrastructure, such as more efficient hardware and optimized deployment pipelines, also play an important role in enabling more sustainable and environmentally responsible use of LLM-enhanced recommender systems.

### III. RESEARCH METHODS

The present study employed an SLR to investigate the integration of LLMs into recommender systems. The SLR method offers a systematic and transparent approach for developing research inquiries, selecting key search terms, and organizing the collected papers in a structured manner. This method enables a comprehensive comprehension of the research subject while maintaining methodological precision and clarity.

#### A. REVIEW PROTOCOL

The research method conforms to the established guidelines set forth by Kitchenham and Charters [88] and Safaei [89],

specifying three key phases: (1) planning the SLR, (2) conducting the SLR, and (3) reporting and disseminating the findings. Research question formulation, development of the review protocol, establishment of inclusion and exclusion criteria, design of the search strategy, outline of the study selection process, and procedures for quality assessment, data extraction, and result synthesis are among the specific tasks involved in each phase. Figure 1 demonstrates the detailed study workflow and protocol.

#### B. FORMULATING RESEARCH QUESTIONS

The subsequent research questions (RQ) were established for directing this SLR:

**RQ1:** What are the current methods or frameworks for integrating LLMs into recommender systems?

**RQ2:** What is the taxonomy and classification of LLMs-enhanced recommender systems?

**RQ3:** How are LLMs currently applied in recommender systems across different domains?

The study questions were formulated to encompass the methodological frameworks, the systematic classification of LLM-based recommender systems, and the various domains in which LLMs are employed for recommendation, thereby providing a comprehensive understanding of the current research landscape in this field.

#### C. SEARCH STRATEGY

A comprehensive, multi-source search approach was employed to identify the integration of LLMs into recommender systems systematically. The Web of Science (WoS) core collection database was chosen as the primary source due to its comprehensive coverage of high-quality, peer-reviewed scholarly literature across several relevant disciplines. WoS enables access to reputable publications and enables comprehensive filtering options for systematic searches. Additional sources consisted of: (i) Scopus, for extended multidisciplinary coverage; (ii) Google Scholar, for the identification of grey literature and preprints; and (iii) manual searches.

The following Boolean query was used in the SLR:

("large language model\*" OR "LLM\*") AND ("recommender system\*" OR "recommender system\*") AND ("cold-start" OR "new user problem" OR "new item problem")

The search was limited to articles published from 2018 to 2025 that included the current introduction and evolution of LLM technologies while preserving historical context. The choice of 2018 as the inception year aligns with the emergence of transformer-based models (GPT) and their use across diverse applications.

#### D. STUDY SELECTION PROCESS

A rigorous multi-stage filtering procedure was used in the study selection process to ensure that only relevant, high-quality studies were included in the final analysis. This procedure consisted of the subsequent phases:

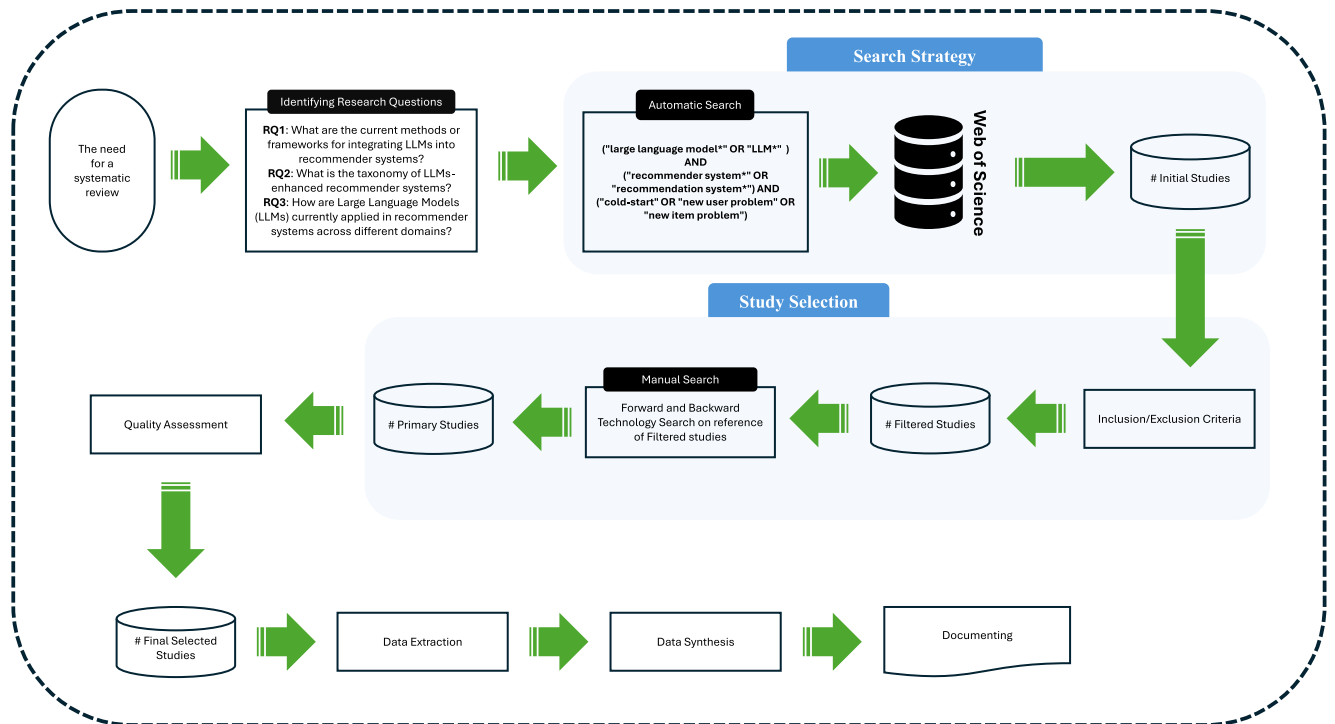


FIGURE 1. Review protocol.

### 1) INCLUSION AND EXCLUSION CRITERIA

Studies were systematically included if they met all of the following criteria: (i) empirical or conceptual studies examined the integration of LLMs or transformer-based models within recommender or recommender systems; (ii) were published in peer-reviewed journals or conference proceedings; (iii) were written in English; and (iv) provided adequate, detailed information regarding implementation or methodology (v) published or made available between January 2018 and December 2025.

Studies were excluded from the review if any of the following criteria applied: (i) The study did not specifically examine or discuss the use of LLMs within a recommender system context; (ii) The publication was an opinion piece; (iii) the study did not demonstrate sufficient methodological rigor or empirical validation; (iv) were duplicates or studies for which the full text was inaccessible.

### 2) MULTI-STAGE FILTERING PROCESS

The selected studies were evaluated by systematic filtering over the three following consecutive stages to ensure an accurate inclusion/exclusion assessment.

(i) Title and Abstract Screening: using focused Boolean queries, we conducted automated searches across WoS, conference proceedings, and Google Scholar. The search was limited to English-language, peer-reviewed research published between 2018 and December 2025, and any study deemed potentially relevant to LLMs–recommender system integration was retained to avoid premature exclusion.

In total, 745 records were screened, of which 189 were selected for full-text review, while the remaining 556 were excluded for irrelevance, language, or publication date.

(ii) Full-Text Eligibility Assessment: We carefully reviewed all 189 full-text articles employing established inclusion and exclusion criteria. Upon evaluating methodological rigor and pertinent relation to LLM–recommender system integration, 74 papers were selected while 115 were removed due to insufficient relevance, lack of empirical rigor, unavailability of full text, or duplication.

(iii) Forward and Backward Citation Tracking: We additionally examined the reference lists and citing articles of the 74 selected studies to identify any works that the original search may have overlooked. This technique found 5 pertinent studies, which were included for full-text review. This increased the total number of studies subject to quality assessment to 79.

### E. QUALITY ASSESSMENT

We used a quality assessment criterion for each selected primary study to assess how well they followed research guidelines and how useful they were to the field. Quality criteria were made to judge: (i) Clarity of research goals and questions; (ii) Methodological rigor; (iii) Data quality; (iv) Validity of conclusions; and (v) Contribution to the field: Does the study advance understanding of LLM integration in recommender systems?

Each study was evaluated using five quality assessment criteria, with scores ranging from 0 to 5 for each criterion,

**TABLE 1.** Experimental quality strengths and limitations of 55 primary studies.

| Quality criteria                   | Key strengths                                                                                                                                                                                                                              | Limitations                                                                                                                             | Included studies                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clarity of research goals          | Explicitly outlined task, model function, and anticipated enhancement (ranking, conversational suggestion, cold-start, explainability).                                                                                                    | Several studies, particularly surveys, encapsulate patterns without empirically testing measurable assumptions through experimentation. | [5], [17], [50], [65], [76], [71], [62], [46], [72], [66], [21], [68], [7], [22], [24], [26], [13], [63], [16], [8], [51], [20], [67], [73], [74], [10], [85], [36], [59], [45], [35], [75] |
| Methodological rigor               | Appropriate baselines, standardized measures, and organized evaluation pipelines. Many studies adopt standard evaluation protocols for ranking and recommendation tasks.                                                                   | Missing details on prompts, hyperparameters, training settings, and ablation studies in some papers.                                    | [5], [17], [50], [65], [26], [13], [63], [16], [8], [51], [20], [73], [74], [10], [85], [24], [66], [62], [72], [68], [67], [7], [22], [76], [21], [16], [71], [51], [36], [66], [62], [72] |
| Data quality and dataset selection | Several works address data sparsity and cold-start problems through augmentation, simulation, or semantic reasoning methods.                                                                                                               | Some studies rely on narrow-domain datasets or synthetic data without sufficient justification or external validation.                  | [17], [50], [65], [13], [8], [24], [26], [85], [35], [59], [45], [20], [73], [74], [10], [5], [71], [62], [7], [22], [76], [21], [13], [16], [71], [51], [36], [66], [62], [72]             |
| Validity of conclusions            | Quantitative improvements documented using standard evaluation metrics (Precision, Recall, NDCG), sometimes supplemented with qualitative analysis.                                                                                        | Numerous studies lack statistical testing, reproducibility validation, or robustness analysis under distribution shifts.                | [5], [17], [50], [13], [8], [26], [85], [24], [66], [62], [72], [66], [76], [21], [13], [16], [71], [51], [66], [68], [67], [20], [73], [74], [10], [71], [62], [7], [22]                   |
| Contribution to the field          | Proposed novel LLM-based recommender approaches including ranking optimization, conversational recommendation, reasoning-based recommendations, knowledge graph integration, retrieval augmentation, and multimodal recommendation agents. | Various conceptual and architectural frameworks lack empirical validation, limiting their readiness for deployment.                     | [65], [62], [17], [50], [26], [85], [67], [20], [73], [74], [10], [66], [59], [45], [7], [22], [76], [21], [13], [16], [71], [51], [66], [72], [68], [36], [71], [62], [24], [66], [8]      |

resulting in a maximum possible score of 25 points per study. Two independent reviewers conducted the evaluation process, and any disagreements were resolved through discussion to ensure consistency and reliability. To be included in the final analysis, studies were required to achieve a minimum total score of 15 points, corresponding to 60% of the maximum score. After applying the quality assessment criteria, 24 articles were excluded due to insufficient methodological rigor, lack of empirical support, or inadequate methodological detail. Consequently, 55 studies met the predefined quality threshold and were retained for data extraction and synthesis. Among the 55 retained studies, the total quality scores range from 21 to 25 out of 25 points. The average quality score across all these 55 studies is 22.8, which significantly exceeds the minimum threshold of 15.

It is important to note that although the selected studies demonstrate sound experimental design and make meaningful contributions to the field, some limitations remain, such as incomplete reproducibility details, limited statistical validation, or insufficient real-world evaluation. Table 1 presents the 55 primary studies selected based on the defined quality assessment criteria; see Appendix: Table 6. Each retained study contributes to one or more aspects of experimental quality, including clarity of research objectives,

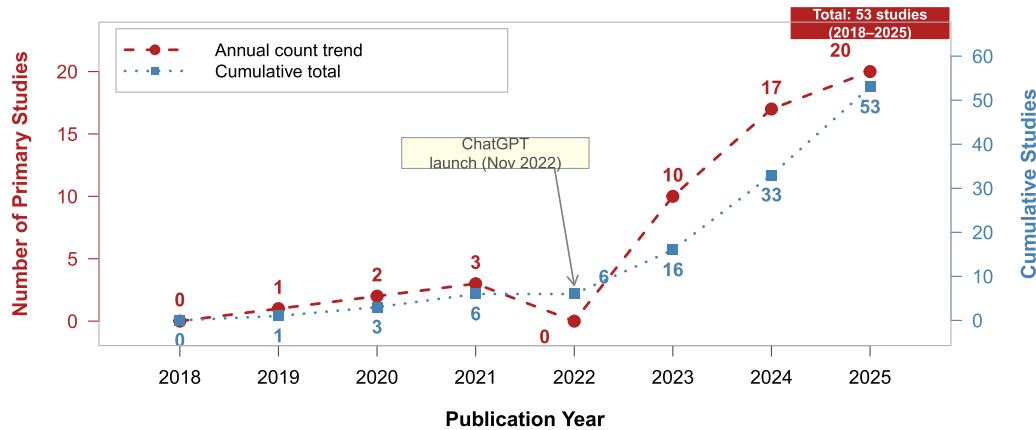
methodological rigor, data integrity, validity of conclusions, and overall contribution to the discipline.

## F. DATA EXTRACTION AND SYNTHESIS

Data extraction was systematically conducted for each primary study. The extracted elements included: (i) study metadata, encompassing author, year, publication venue, and study type; (ii) the research focus and stated objectives; (iii) the LLM models and architectures employed; (iv) the types of recommender systems examined; (v) the specific application domains in which the recommender systems were deployed; (vi) methodological approaches, including implementation frameworks and analytical procedures; (vii) evaluation metrics and reported performance outcomes; and (viii) key findings along with identified limitations. This systematic extraction method ensured consistency among all examined studies and enabled a thorough synthesis of the research landscape. Data elements defined in Table 1.

### 1) PUBLICATION TIMELINE ANALYSIS

An examination of the 2018-2025 references identifies eight unique research focal points, each signifying significant research trends and application domains within LLM-enhanced recommender systems, as illustrated in Figure 2.



**FIGURE 2.** Publication timeline of 55 primary studies on LLM-Enhanced recommender systems (2018-2025).

**TABLE 2.** Data extraction schema.

| Data Element            | Description                                          |
|-------------------------|------------------------------------------------------|
| Study Metadata          | Author(s), publication year, venue, study type       |
| Research Focus          | Main objectives and research questions               |
| LLM Models              | Specific LLM architectures and versions used         |
| Recommender System Type | Collaborative filtering, content-based, hybrid, etc. |
| Application Domain      | E-commerce, healthcare, media, social networks, etc. |
| Methodology             | Implementation frameworks and technical approaches   |
| Evaluation Metrics      | Performance measures and experimental setup          |
| Key Findings            | Main results, contributions, and limitations         |

The distribution indicates a distinct change from singular-method strategies to comprehensive, multi-modal solutions; see Table 2. Research on LLM-Based recommender systems are grouped in ten themes as shown in Table 3 from LLM integration methods to explainability & interpretability of LLM in recommender systems.

## 2) MAJOR PUBLICATION VENUES AND RESEARCH COMMUNITIES

The 2018–2025 references indicate that significant research activity is concentrated within a few key publication venues. ArXiv accounts for approximately 30% of all studies, reflecting the rapid pace of innovation and the community's preference for swift dissemination of emerging ideas. ACM Transactions on Information Systems (TOIS) contributes about 15%, while major conferences such as ACM CIKM and SIGKDD collectively represent around 20%, demonstrating strong engagement from the core information-retrieval and data-mining communities. IEEE journals publish roughly 7% of the studies, and an additional 9% appear in domain-specific journals. Specialized journals contribute about 17%,

highlighting the growing applied relevance of LLM-enhanced recommender systems across diverse sectors. Only 2% of the publications appear in ACM Computing Surveys, indicating that large-scale integrative reviews are still emerging. Overall, this distribution underscores both swift innovation and increasing sophistication in research on LLM-enhanced recommender systems.

## IV. RQ1: WHAT ARE THE CURRENT METHODS OR FRAMEWORKS FOR INTEGRATING LLMs INTO RECOMMENDER SYSTEMS?

Numerous studies and experiments have been conducted to investigate the integration of LLMs into recommender systems. Collectively, these efforts reveal a clear shift toward incorporating LLM-driven semantic reasoning, contextual understanding, and natural language interaction into traditional RS components. Figure 3 presents a framework for integrating LLMs into recommender systems, illustrating how LLM capabilities can be embedded across different stages of the recommendation pipeline.

### A. CLLM4REC FRAMEWORK

The CLLM4Rec framework has been proposed, which stands for Collaborative Large Language Model for Recommender Systems. This framework synthesizes and utilizes LLM with the ID paradigm of RSs. To illustrate, the traditional ID-based approach will be integrated with an LLM to expand its vocabulary, enabling it to capture collaborative and semantic content [76].

**A prompting structure** A prompting structure has been introduced to break down into soft tokens represent user and item IDs that are learned as continuous embeddings and Hard Tokens are concepts from natural language which enable the LLM understand semantic attributes, contextual descriptions, and text [51] This separation improves the LLMs overall effectiveness by allowing the LLM components contribute to the final prediction task to specialize in both content-based and collaborative filtering methods [76]. Zhu et al.



TABLE 3. Summary of research themes in LLM-based recommender systems.

| Research Category                    | Counts | Period | References                                                     |
|--------------------------------------|--------|--------|----------------------------------------------------------------|
| LLM Integration Methods              | 11     | 22–25  | [5], [11], [69], [12], [76], [68], [19], [77], [6], [21], [22] |
| Cold-Start Problem Solutions         | 7      | 14–25  | [13], [16], [65], [62], [63], [42], [4]                        |
| Prompt Engineering & Fine-tuning     | 6      | 21–25  | [71], [51], [72], [70], [35], [82]                             |
| Conversational & Interactive Systems | 6      | 20–23  | [66], [67], [60], [34], [36], [37]                             |
| LLM as Ranker/Retriever              | 5      | 24–25  | [11], [73], [74], [79], [69]                                   |
| Data Augmentation & Quality          | 5      | 24–25  | [15], [65], [85], [14], [23]                                   |
| Domain-Specific Applications         | 6      | 23–25  | [80], [81], [83], [54], [45], [90]                             |
| Knowledge Graphs & Multimodal        | 4      | 19–24  | [17], [75], [91], [84]                                         |
| Hallucination & Robustness           | 3      | 23–23  | [24], [25], [55]                                               |
| Explainability & Interpretability    | 2      | 20–24  | [26], [39]                                                     |

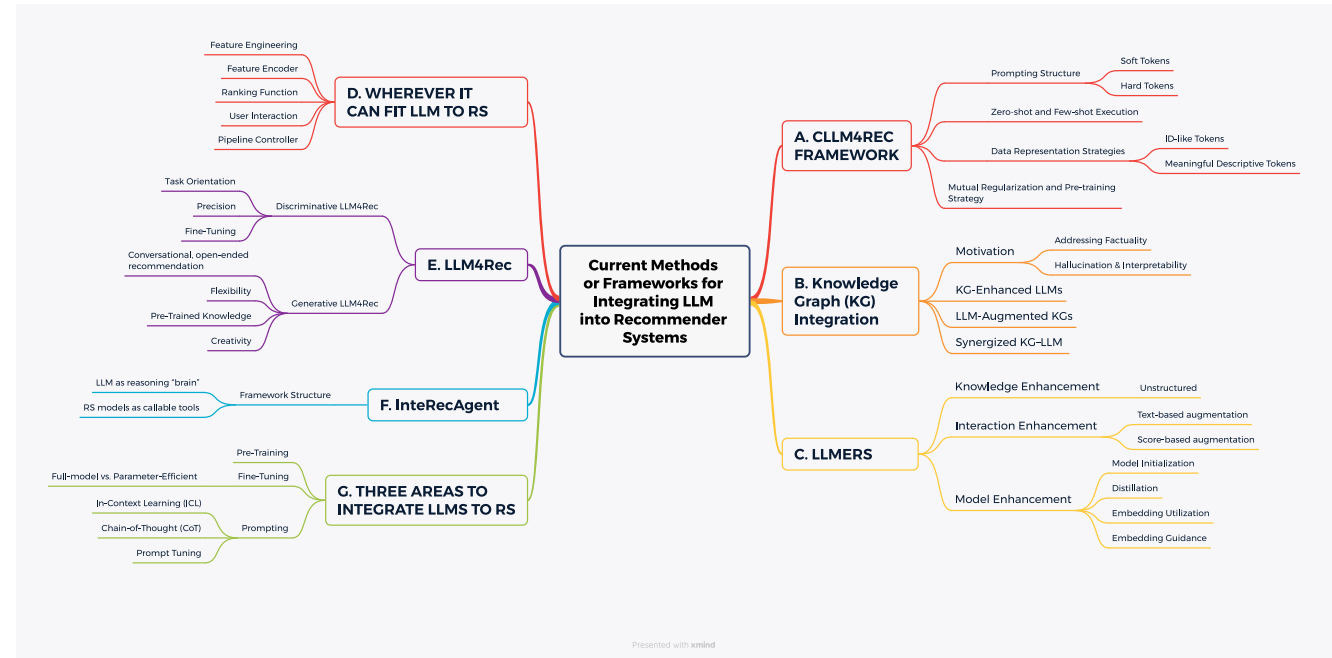


FIGURE 3. Framework for integrating LLMs in recommender systems.

[76] present CLLM4Rec, which creates high-quality prompts based on user interests, behaviors, platform interactions, and item characteristics. These prompts are then fed to LLM models to generate final recommendations. This process can be executed using a zero-shot approach [5], [11], [60], [69]. The framework employs a soft and hard prompting strategy that successfully pre-trains CLLM4Rec on tokens encoding historical interactions as well as user and item features. This method utilizes mutual regularization to enable the model to efficiently capture both collaborative and content information through user and item token embeddings [51], [76].

It is worth noting that user and item data points can be represented in two primary ways: ID-like tokens and semantically meaningful tokens. In the first approach, an identifier is constructed using a prefix such as “product”

or “user”, followed by an underscore, and a unique numeric value. This strategy is practical for real-world systems; however, during tokenization, the identifier may be split into multiple sub-tokens—for example, “product\_34159” may become [”product”, “\_”, “341”, “59”]. In contrast, the second approach relies on descriptive and meaningful words to represent users or items. While this can enrich semantic understanding, it may also introduce biases in user–item similarity due to unintended semantic associations [11], [12].

**Zero-shot and Few-shot Execution** Leveraging this prompting methodology, CLLM4Rec excels in scenarios with inadequate labeled training data. In the most basic scenario—termed zero-shot mode by researchers—the LLM can leverage its pre-training skills and utilize the structured prompts derived from user histories and item characteristics

to produce suggestion ranks immediately, without requiring fine-tuning [5], [11], [69]. When you have several annotated examples, you can shift into few-shot mode with relative ease. In this instance, incorporate several concrete examples—such as particular user-item pairs accompanied by explanations directly into the prompt to facilitate improved rating behavior by the LLM. This method aligns with increasing evidence that LLM-based recommenders can function as remarkably excellent zero- and few-shot rankers, demonstrating effectiveness in conversational recommendation across several domains [10], [11], [19], [60].

**Data Representation Strategies** Within this framework, a significant decision you will encounter is the method of encoding user and item information. As previously stated, there are fundamentally two primary approaches in this context. You may opt for ID-like tokens, such as “user\_1234” or “product\_34159,” or select meaningful descriptive tokens that include semantic features, such as “family-budget-traveller” or “impressionist-art-museum” [11], [12]. Each approach has its trade-offs. ID-like tokens are great at identifying collaborative filtering signals and work effectively with traditional recommender system architectures. But there’s a problem: during BPE-style tokenization, they often get chopped up into several sub-tokens, which can make the interaction patterns you’re trying to encode less effective. On the other hand, descriptive tokens enable you use the LLM’s built-in semantic knowledge, which is quite helpful when you have sparse data or cold-star items. But they have their own problems: terms that are similar in meaning can unwittingly mess up your similarity relationships by putting too much emphasis on specific user or item features [4], [16], [42], [62].

**Mutual Regularization and Pre-training Strategy** Lastly, CLLM4Rec trains the system in a smart way: instead of optimizing the ID-based recommender and the LLM component individually, it trains them together using a method called mutual regularization [76]. This is how it works. The soft-token channel, which includes your user and item ID embeddings, learns to spot high-order collaborative patterns in huge logs of interactions. The hard-token channel, on the other hand, deals with natural-language descriptions, reviews, and other side information. It is more concerned with semantic reasoning and coming up with explanations [33], [51].

## B. KNOWLEDGE GRAPH (KG)

The Knowledge Graph, introduced by Google in 2012, could be a solution if it integrates with LLMs, as it utilizes a graph-based model to manage and leverage information from multiple sources on a large scale [91]. There are two main **Motivation** reasons for integrating KG into LLMs based recommender systems: to improve hallucination control and interpretability and to make sure that the facts are correct [24]. LLMs are good at reasoning, but they often have trouble with factual reliability because their knowledge is stored in parameters rather than in clear structures. This makes it hard

to find correct or domain-specific relationships, especially for long-tail items [17], [49]. KGs solve this problem by using factual information to ground LLM reasoning and make recommendations for more reliable and precise [17], [91]. LLMs are also prone to hallucination, which means they can make up plausible but incorrect descriptions or user preferences [24], [55]. KGs help with this problem by acting as an external organized memory that limits generation and offers clear relational paths, making it possible to give explanations that are consistent with explicit graph reasoning. This integration makes things clearer, more trustworthy, and easier to understand—all of which are important for real-world recommender systems [26], [39].

This KG and LLMs combination has been implemented in a wide range of real-world applications for advice, question answering, recommendation, and web search purposes [17]. The integration methods include: **KG-enhanced LLMs** pre-training, inference, and interpretability stages, which enhance LLMs’ knowledge expression, update them to the latest knowledge without retraining, and interpret their reasoning process. **LLM-augmented KG** encompass tasks such as embedding, completion, construction, text generation, and question answering. Enhancing KGs representations, generating factual information for better KG performance, addressing difficulties in entity discovery, creating natural language factual text, and eventually bridging the gap between natural language questions and answer retrieval from KGs. **Synergized KGs and LLMs** Synergized KGs and LLMs are a two-way integration in which each part improves the other. KGs give LLMs structured, factual information to minimize hallucinations and make them more reliable. LLMs, on the other hand, add semantic reasoning to the KG to make it more detailed, more complete, or more precise. They work together to form a single system that blends symbolic accuracy with neural understanding to give more accurate and understandable recommendations [17].

## C. LLMERS

The total time it takes for LLMs to receive a user’s prompt and generate a complete response can be a few seconds, while RSs typically require faster responses and lower latency [7]. The term LLMERS refers to the integration of LLMs into RSs, which can be grouped into three types: **Knowledge Enhancement** leverages the world knowledge and reasoning capabilities of LLMs to generate text for users and items. Prompts may be initially provided by users and then processed to create unstructured summary text, structured knowledge graphs, or a combination of both, enabling LLMs to generate user preferences and ultimately improve the efficiency and accuracy of RSs. In addition, to address the data sparsity issue, data augmentation is an effective method [14] for creating new interactions through LLMs, which can then be used to train traditional RSs. These RSs can be divided into two output groups: text-based and score-based.

The first group generates additional user–item interactions, such as recommendations and inferred preferences, while simultaneously removing noisy data. The second group augments data based on similarity and confidence measures. This approach is referred to as **Interaction Enhancement** [7], [65]. Moreover, through **Model Enhancement**, LLMs can analyze interactions from a semantic perspective [7]. RS models consist of two parts: an embedding layer that captures item relationships and hidden layers that extract user preferences. This integration can be divided into Model Initialization, Model Distillation, Embedding Utilization, and Embedding Guidance. Specifically, the first two improve the entire RS model, while the last two focus on the embedding layer. Additionally, Model Initialization, Model Distillation, and Embedding Guidance are used solely during training and are not accessible during inference [7].

#### D. WHEREVER IT CAN FIT LLM TO RS

In the RS pipeline, there are several stages where LLMs can be applied, including feature engineering, feature encoding, ranking functions, user interaction, and pipeline control. By using LLM for **Feature Engineering**, the original features such as item and user data provided as input can be transformed, and new features can be generated to improve user and item understanding while augmenting data for training [65]. LLM for **Feature Encoder** introduce semantic information for downstream neural RSs, enrich user and item representations, and support cross-domain recommendations [76]. LLM for **Ranking Function** offer a direct way for LLMs to act as recommenders by estimating utility scores for user-item pairs or generating probability distributions, producing ranked item lists. A hybrid alternative uses multitask prompting to both compute ranking scores and generate ranked items [73]. LLM for **User Interaction** enable a dialogue-oriented approach in which users interact with the system conversationally, allowing the model to more clearly identify user intentions and preferences and, as a result, offer improved recommendations. Finally, LLM for **Pipeline Controller**, the entire recommendation workflow can be orchestrated by the LLM, functioning as an agent capable of reasoning, calling APIs, performing calculations, and utilizing third-party tools and scripts [22].

#### E. LLM4REC

LLM4Rec architecture is grouped by their use-cases into two types of discriminative and generative paradigms. **Discriminative LLM4Rec** systems take the given input data and match items and products that align with users' preferences [20]. This method has several key characteristics: (1) "Task Orientation" to perform specific actions such as predicting the next item or personalizing content, (2) "Precision" to achieve high-accuracy recommendation results. and (3) "Fine-Tuning" to optimize performance within particular domains [11], [77]. On the other hand, **Generative LLM4Rec** systems are well suited for conversational agents and chatbots, where interactions can continue

with open-ended responses. This approach also has several characteristics: (1) "Flexibility" enabling the handling of diverse actions such as dialogue-based interaction and open-ended suggestions, (2) "Pre-Trained Knowledge" which supports few-shot and zero-shot learning, and (3) "Creativity" allowing the system to generate innovative and enriched responses. As a result, combining these two paradigms within the LLM4Rec architecture provides complementary strengths and advantages, leading to more effective solutions for different recommendation scenarios [8].

#### F. INTEGRATED RECOMMENDER AGENT (INTERECAGENT)

Conversational Recommender Systems are segmented into two types: (1) open-ended conversation, which utilizes pretrained language models to understand dialogue and generate responses, and (2) attribute-based question–answering, which interacts back and forth with users to refine their interests over time and recommend suitable options [66]. However, these approaches are not highly functional in practice and often struggle to address complex user questions or maintain coherent conversational flow. This limitation motivated the development of the **InteRecAgent framework** proposed by [68], which transforms traditional recommender systems into interactive systems powered by LLMs. In this framework, the LLM serves as the "brain" of the system, while recommendation models act as tools that supply domain-specific knowledge. As a result, the LLM receives user inputs and then decides whether it needs to call a tool or script to complete the task, or whether it can generate the response using its own internal knowledge [68].

#### G. THREE AREAS TO INTEGRATE LLMs TO RS

LLMs can be integrated in RSs in three ways: (1) **Pre-Training** involves training LLMs on a comprehensive amount of textual data to improve their overall understanding and enable strong generalization and zero-shot prompting capabilities. (2) **Fine-tuning** optimizes the model for specific types of tasks and allows it to grasp domain knowledge. This process is divided into two categories full-model fine tuning and parameter-efficient fine tuning which differ in computational cost and the proportion of model weights updated during training. (3) **Prompting** converts various tasks into text generation tasks and is considered a lightweight and cost-effective alternative to pre-training and fine-tuning. Through prompting, recommendation tasks and other RS operations can be framed as language generation problems [22], [51]. Within the prompting stage, several techniques are applied, including in-context learning (ICL), Chain-of-Thought (CoT), and Prompt Tuning. In ICL, LLMs learn new or unseen tasks from examples included within the prompt during inference, enabling both few-shot and zero-shot performance. Few-shot typically outperforms zero-shot due to the additional input and explanations provided. Moreover, CoT prompting allows LLMs to break down complex decision-making processes and produce the final result through step-by-step reasoning [19].

**TABLE 4.** Comparative analysis of major LLM integration frameworks in recommender systems.

| Framework     | Role of LLM                 | Main Function                                   | Strengths                               | Limitations                               |
|---------------|-----------------------------|-------------------------------------------------|-----------------------------------------|-------------------------------------------|
| CLLM4Rec      | Collaborative hybrid        | Joint semantic and collaborative recommendation | Strong performance, cold-start handling | Tokenization and prompt design complexity |
| KG-Integrated | Knowledge reasoning engine  | Grounded reasoning with structured knowledge    | High accuracy, reduced hallucination    | Knowledge graph maintenance required      |
| LLMERS        | Enhancement module          | Improves existing recommender components        | Backward compatible                     | Computational overhead                    |
| LLM4Rec       | Core recommender model      | Ranking or generative recommendation            | Flexible, conversational support        | Hallucination risk                        |
| InteRecAgent  | Autonomous agent controller | Orchestrates tools and recommendation workflow  | Highly intelligent and adaptive         | High complexity                           |

These frameworks outline a spectrum of integration advancements. The pipeline-level and LLMERS frameworks offer a set of useful improvements over current systems, while CLLM4Rec and LLM4Rec offer more advanced semantic reasoning and generative capabilities. Knowledge Graph-based frameworks enhance factual knowledge grounding. Agent-based frameworks such as InteRecAgent represent the future of autonomous, reasoning-based recommendation systems. Table 4 summarizes comparative analysis of major LLMs integration in recommender systems. Overall, these frameworks illustrate the revolutionary potential of LLM integration to enhance recommendation accuracy, personalization, interpretability, and adaptability in various application areas.

## V. RQ2: WHAT IS THE TAXONOMY AND CLASSIFICATION OF LLMs- ENHANCED RECOMMENDER SYSTEMS?

Previous studies on the role of LLMs in RS can be categorized based on: (A) Role of LLMs in the RS Pipeline, (B) Cold-Start Mitigation Strategy, (C) LLM Integration Approach, (D) System Architecture, and (E) Evaluation Dimensions. These categories are illustrated in Figure 4 and outlined in the following subsections.

### A. ROLE OF LLMs IN RS PIPELINE

The role of LLMs in the RS pipeline taxonomy highlights how LLMs improve deep learning-based RS and participate in various phases of RS by having essential responsibilities in user data acquisition, feature engineering, and the ranking process. LLMs surpass their function as simple components by actively managing the system pipeline, promoting participatory and transparent recommender systems [21]. The research examined in the SLR divides the importance of LLMs in the RS pipeline into three primary domains: (i) content representation, (ii) interaction comprehension, (iii) generation and summarization.

**Content Representation** in RS pertains to the expression of user and item information in a format comprehensible to machines, enabling algorithms to compare, match, and produce recommendations. LLMs provides advanced functionalities for recommender systems by revealing

complicated semantic relationships [8]. Semantic relationships derive meaning from textual content, including user reviews, item descriptions, and metadata.

In their study on content representation, Liu et al. [7] explained the role of LLMs in the RS pipeline in the following three folds. Knowledge enhancement emphasizes the ability of LLMs to improve system inputs, including the generation of high-quality semantic embeddings for users and items, as well as the augmentation of knowledge graphs with external information. Interaction enhancement facilitates context-sensitive recommendations derived from natural language inquiries. This enables conversational agents and LLMs to recognize and react to users, such as by converting natural language queries into suggestions or facilitating conversational recommendation agents. Model enhancement entails the integration of LLMs directly into the recommender model to help with ranking, rating, or inference procedures.

Lin et al. [22] investigate the role of LLMs in RS and analyze the integration of LLMs throughout the whole RS pipeline, particularly at the feature engineering step, feature encoding stage, in the stage of scoring or ranking, as well as in the user interaction stage, and pipeline management stages. They highlighted that LLMs significantly improve feature representation via semantic embeddings, facilitating precise encoding of textual content, including user reviews, item descriptions, and metadata. This enhancement boosts the capacity of the system to represent both user preferences and item characteristics. Furthermore, they also asserted that LLMs enhance interaction comprehension by facilitating conversational suggestion contexts and natural language inquiries, thus aligning user intent with system responses.

This aligns with related work, such as Pan et al. [17], and [50], which further illustrate how LLMs can incorporate external knowledge and reasoning capabilities into RS. These studies collectively affirm that knowledge enhancement via content representation is a fundamental function of LLMs within the RS pipeline.

**Interaction Comprehension** in LLMs refers to their ability to understand the purposes, complexity, and dialogues of human-machine interactions. Moreover, it enables LLMs



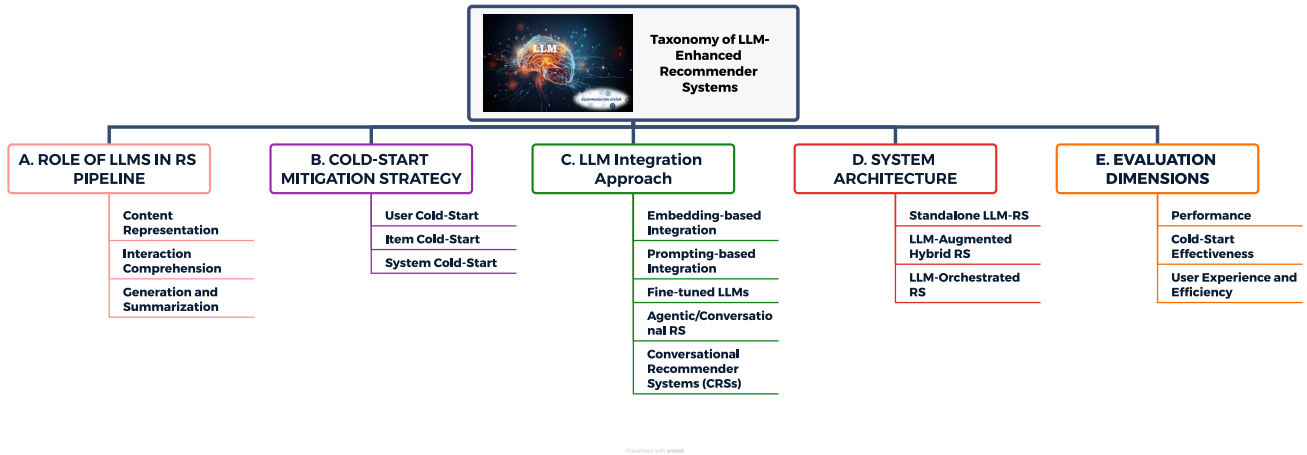


FIGURE 4. Taxonomy of LLM-Enhanced recommender systems.

to evaluate natural language inquiries, comprehend user inputs, and provide conversational recommendations. This thereby links human-like dialogue with recommendation results.

LLMs' role in RS typically engages in multi-turn dialogues. By incorporating context into conversations and utilizing the vast amount of open-world knowledge, LLM offers an unparalleled depth of understanding of users' needs and interests throughout such a dialogue Lin et al. [22]. A successful recommendation fundamentally depends on a precise understanding of the user's underlying demands. Zhang et al. [69] emphasized the necessity of formulating an appropriate structure to thoroughly articulate the recommendation task in natural language, including numerous variables such as interaction history, user preferences or intentions, and more personalized aspects.

In addition to representation and interaction, LLMs also facilitate **Generation and Summarization**. This includes providing natural language explanations, ensuring interpretability, and summarizing evaluations or feedback from users. Xu et al. [26] asserted that LLMs enhance RS by producing natural language explanations. Thus, it augments transparency and interpretability. Based on this, generation and summarization help RS turn enormous amounts of text data, such as user reviews or product details, into short, valuable insights. Sun et al. [58] enhance this viewpoint by demonstrating how summarization and explanation can reveal hidden biases, thus endorsing more equitable and trustworthy recommendations. Their research indicates that LLMs can clarify outcomes of models and identify potential equitable concerns in the recommendation process.

### B. COLD-START MITIGATION STRATEGY

The cold-start problem is a big concern for recommender systems. This problem occurs when a recommender system can't make accurate suggestions due to insufficient historical data [4]. This problem shows up in three key ways in

recommender systems. (i) User Cold-Start, (ii) Item Cold-Start, and (iii) System Cold-Start.

The **User Cold-Start** problem is one of the significant challenges in RS. When a new user signs up on the platform, the system lacks information about prior interaction history and behavior patterns, which makes it hard to offer recommendations based on user preferences [43]. The user cold-start problem can be minimized by using techniques such as preferred selection and adaptive learning based on user interactions and prompts [23].

**Item Cold-Start** is a challenge in recommender systems where a new item is introduced to the system without sufficient user interaction data, making it challenging to recommend effectively [42]. LLMs are becoming powerful tools to help with this issue by providing novel methods of demonstrating and integrating unseen items into the recommendation pipeline. To address the cold-start challenge in RS, a study by Rungratnont and Mongkolnavin [63] presents a novel method that combines data augmentation with cold-start item search strategies with the interpretative power of LLMs. The system is able to identify user preferences and provide pertinent recommendations for both warm and cold items when using the LLM search method. These discoveries further the development of RS, particularly in resolving cold-start issues.

Huang et al. [13] proposed a new LLM Simulator framework to solve the item cold-start problem at its core. They developed ColdLLM, which addresses the cold-start problem in large recommender systems, enabling a significant improvement in performance and having a bigger effect on the economy. The observation suggests that ColdLLM creates new opportunities for using LLMs in extensive online RS. Therefore, LLMs can employ textual features and extensive pretraining knowledge to help reduce the cold-start issue for new users or items. LLMs may generate significant representations for cold-start users and items, thus enhancing the quality of recommendations.

The **System Cold-Start** issue in RS systems arises from a lack of training data, which hinders their ability to make efficient recommendations. To address the system cold-start issue, Zhang et al. 2024 [16] presented LLMTreeRec, an LLM-centered tree-based recommendation architecture. They present LLMTreeRec, a tree-based LLM recommendation framework. They developed a unique method to organize all items into a hierarchical tree structure, or item tree, in order to handle large item sets. LLMs use this hierarchical structure for search in order to efficiently refine the candidate item set based on the item tree.

Wu et al. 2024 [62] in their studies examined the cold-start recommendation problem and presented the initial benchmark. They developed a basic, however effective, in-context learning method employing language models, named PromptRec. These language models could address this issue and demonstrate that LLMs can provide specific recommendations without requiring any training examples.

### C. LLM INTEGRATION APPROACH

**Embedding-based Integration** method involves encoding node texts and their adjacent structural information into compressed embeddings, which are subsequently analyzed by LLMs [75]. Since LLMs are not intrinsically trained on random embedding spaces, it is crucial to correlate these embeddings with the LLM's token space—a concept partially derived from the alignment of multimodal data in vision-language models (Radford et al. [99], Zhai et al. [100]). In contrast to the vision-language domain, which possesses an abundance of large-scale text-image pairs (Schuhmann et al. [101]), the graph domain generally lacks analogous datasets. This shortage diminishes the model's practical generality.

**Prompting-based Integration** is a prominent approach for optimizing the effectiveness and precision of LLMs in certain natural language processing applications, and it is also considered a crucial method for human-computer interaction. It facilitates the explicit communication of clear task descriptions to LLMs, which then provide replies aligned with user expectations through analogical learning [71]. As the LLMs capability increases, prompts created for certain specialized tasks are typically more varied and comprehensive. Thus, developing appropriate prompts can enable LLMs to recognize user desires and deliver recommendations with intended understanding and knowledge application abilities [72].

A robust teacher-LLM (GPT-3.5) was utilized by Zhang et al. [69] to produce individualized information for each user by utilizing their history interactions and reviews, which provide comprehensive insights into desires and intents. By using LLMs with step-back prompt approaches, Zeng et al. [70] proposed RoSiLC-RS, which helps the model comprehend legal concepts and principles at a higher level. This enables the model correctly identify important aspects of similar instances in vast amounts of textual information.

Moreover, Mao et al. [72] in their study offered instance-wise prompting, indicating the generation of customized prompts for each individual user. They proposed Reinforced Prompt Personalization (RPP) for automated execution. Rather than constructing prompts incrementally, RPP operates at the phrase level to enhance both speed and quality. It segments a stimulus into four patterns, modifies them utilizing various agents, and after integrates them. These customized prompts undergo regular testing with the LLM, and the feedback enhances the LLM's recommendation efficacy.

**Fine-tuned LLMs** can handle a wide range of downstream needs without needing a lot of reprogramming. This means that a single base model can be used for multiple analytical tasks within the same field, such as predicting load or restoring missing data. This flexibility ensures peak performance even when large, detailed datasets are unavailable due to privacy, security, or practical reasons [83].

Recent developments in LLMs have led to the creation of **Agentic/Conversational RS** that surpass single models iner than single models at doing certain tasks. These agentic systems demonstrate greater independence and flexibility in complex tasks by allowing LLMs to perceive external environments, gather information from multiple sources, and utilize a range of tools. This development opens up new possibilities for RS: LLM-based Agentic recommender systems can offer more interactive, context-sensitive, and proactive suggestions, which could change how users interact with recommender systems and make them more useful in more situations [18].

**Conversational Recommender Systems (CRSs)** have emerged as vital instruments for decision-making. It typically involves several subtasks, such as obtaining user input, making suggestions, explaining concepts, and gathering information about items [66]. To make CRS functions more effectively, we need to address several issues. First, CRSs require a manager for subtasks. We need to address several problems: First, the CRSs require a manager for subtasks. Feng et al. [67] indicated that because conversational recommendation involves numerous subtasks within a coherent dialog interface, the CRSs must be able to determine when to perform each one and when to do so. The CRSs need to be able to handle different subtasks independently. To improve performance, the CRSs need to create and optimize for specific subtasks. The CRSs also need to generate language more effectively. The CRSs must provide users with answers that are sensible, consistent, and relevant.

### D. SYSTEM ARCHITECTURE

It is possible to categorize LLM-enhanced recommender systems based on their reliance on traditional recommendation elements and the LLM's role within the system. **Standalone LLM-RS** operate as independent recommender systems, with the LLM managing the entire recommendation process—interpreting user intent, creating representations, and providing item suggestions without relying on additional

neural modules. These systems leverage in-context learning, prompt engineering, or fine-tuning to quickly infer what people like from natural language data, such as reviews, questions, or interaction history [18]. This architecture focuses on generalization and interpretability, making it easier to do recommendation tasks with zero or a few data points. However, the LLM's domain alignment and lack of systematic user-item interaction modeling may limit performance.

On the other hand, **LLM-Augmented Hybrid RS** adds LLMs as enhancement modules to traditional pipelines. This combines the reasoning and semantic understanding of language models with collaborative filtering, graph-based, or content-based methods. In these systems, the LLM serves as a feature generator, re-ranker, or explanation engine, improving user and item representations to make them easier to understand and resolving cold-start problems [20]. Ultimately, **LLM-Orchestrated RS** showcases an advanced architectural layer, with the LLM serving as an intelligent controller, or “cognitive orchestrator.” It integrates specialized models or APIs, gathers data, activates sub-recommenders, and analyzes intermediate outputs to provide context-sensitive, multimodal recommendations [68].

### E. EVALUATION DIMENSIONS

We can examine LLM Enhanced RS in four primary ways: performance, cold-start efficacy, user experience, and efficiency. For **Performance** evaluation, traditional accuracy-based metrics, such as accuracy, precision, recall, and normalized discounted cumulative gain (NDCG), remain essential. Recent research demonstrates that LLM-based recommenders achieve competitive precision while exceeding conventional models in their capacity to capture semantic and contextual nuances in user-item interactions [8], [19]. By leveraging instruction tailoring and fine-grained ranking adaptation, LLMs can readily be integrated into personalized recommendation tasks. This makes ranking more consistent and easier to understand than traditional deep learning methods [73].

**Cold-Start Effectiveness** investigates how LLM-RS handles situations with new users or new items, which have long been challenging for collaborative filtering and matrix factorization algorithms [74]. In these situations, LLMs demonstrate significant improvement by leveraging extensive textual metadata, user evaluations, and retrieval-augmented generation to identify preference indicators, even when limited behavioral data is available. The combination of semantic reasoning and natural language understanding may allow LLMs to generalize over sparse data contexts. This reduces the need for data and makes it easier to adapt in rapidly changing fields, such as e-commerce and media recommendation [10].

**User Experience and Efficiency** are key factors to consider when evaluating the usability and scalability of a system. Explainability, transparency, and user trust are all key components of the user experience that conversa-

tional reasoning and natural-language justification features in LLMs enhance [26], [82], [85]. At the same time, efficiency is becoming increasingly important, as LLM-based designs are computationally intensive. Studies have examined improvements in latency reduction, scalability, and resource optimization through model distillation, multimodal compression, and fine-tuning techniques [15], [77].

Beyond system-level performance metrics, evaluating recommender systems from the user's perspective is becoming increasingly important. Measures such as perceived recommendation quality, user satisfaction, task completion rates, and A/B testing in real-world environments provide valuable insights into how these systems perform in practice.

However, most studies in the existing literature still rely primarily on offline evaluation methods. As a result, the use of standardized user-centered evaluation protocols remains limited. Addressing this gap represents an important direction for future research, particularly for understanding the true impact of these systems in real-world settings.

## VI. RQ3: HOW ARE LLMs CURRENTLY APPLIED IN RECOMMENDER SYSTEMS ACROSS DIFFERENT DOMAINS?

Given that LLMs have demonstrated remarkable adaptability and effectiveness in managing complex recommendation scenarios by leveraging both structured and unstructured data, this section highlights several domains in which RSs have been successfully integrated with LLMs [8]; see Table 5. Moreover, Figure 5 provides a visual overview of how LLM capabilities are utilized across these domains, followed by detailed discussion in the following subsections. The heatmap uses a four-level scale, ranging from Not Applicable to High, to allow clear comparison of how strongly each capability is applied within different domains. Among the evaluated capabilities, personalized recommendation emerges as one of the most prominent factors, receiving high ratings across most domains.

### A. E-COMMERCE & RETAIL

In the fields of e-commerce and retail, LLMs are extensively employed to determine user intent from various sources, including reviews, search queries, and browsing patterns. This application facilitates the generation of more precise tailored recommendations and the implementation of adaptive customer journey analysis, as demonstrated in prior research [30], [69].

### B. NEWS & CONTENT CURATION

In news recommendation, models like *T5* demonstrate strong proficiency in distilling long articles into clear, concise summaries, helping readers quickly grasp the main points before diving deeper [92]. On the other hand, the rise of fake news, especially with the advent of LLMs, has intensified the global challenge of misinformation and enabled malicious actors to produce large volumes of fake content with ease and

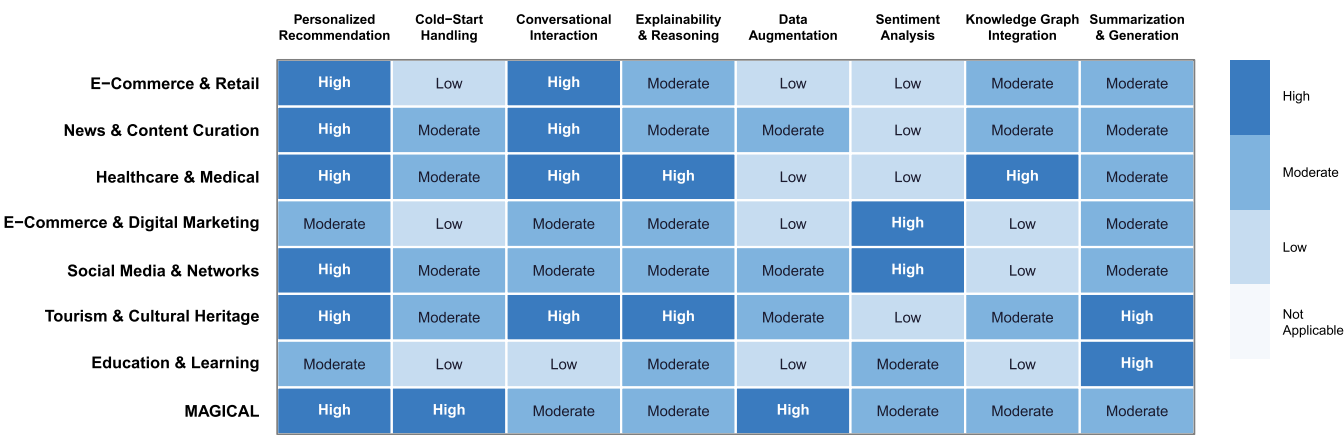


FIGURE 5. Domain application Heatmap of LLM capabilities in recommender systems.

TABLE 5. LLM applications across different domains.

| Domain                         | LLM Capabilities                                                                                       | Key References               |
|--------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------|
| E-Commerce & Retail            | User intent inference; Personalized recommendations; Customer journey analysis; Marketing optimization | [30], [69], [76], [16], [65] |
| News & Content Curation        | Summarization; Key point extraction; User preference modeling; Content ranking                         | [5], [39]                    |
| Healthcare & Medical           | Clinical decision support; Health record synthesis; Outcome prediction; Behavioral coaching            | [53], [54], [25], [55], [24] |
| E-Commerce & Digital Marketing | Sentiment analysis; Behavior trend detection; Dynamic pricing; Retention optimization                  | [30], [67], [85], [68]       |
| Social Media & Networks        | Sentiment inference; Personalized content feeds; User modeling; Preference evolution tracking          | [82], [18], [60], [34]       |
| Tourism & Cultural Heritage    | Context-aware museum guidance; Cultural path recommendation; Interactive visitor engagement            | [80], [81], [97], [95], [83] |
| Education & Learning           | Adaptive tutoring; Course recommendations; Interactive learning support                                | [45], [54], [73]             |
| MAGICAL                        | Utilizing in museums; enjoyment and satisfaction of visitors                                           | [97] [95]                    |

at low cost. Consequently, news RSs on online platforms now face increased credibility risks [92].

C. HEALTHCARE & MEDICAL

In health and wellness, LLMs facilitate personalized recommendations by synthesizing data from electronic health records, wearable devices, and user-provided inputs [8]. LLMs enable healthcare personalization by recommending tailored diet plans, exercise routines, and mindfulness activities [46]. Furthermore, they process clinical notes and structured health data to generate treatment recommendations and actionable insights for both patients and healthcare providers [53]. Moreover, when integrated with behavioral science frameworks, LLMs can provide effective conversational coaching to improve healthier lifestyles through personalized activity recommendations and empathetic dialogue [54].

D. E-COMMERCE & DIGITAL MARKETING

A large quantity of unstructured data, such as product reviews, descriptions, and customers’ reviews, can be analyzed to extract valuable insights into user preferences. LLMs can mine customer reviews to uncover the sentiments and emotions expressed in the text. Moreover, LLMs can enable e-commerce to detect trends in big data and make necessary

changes to marketing campaigns and inventory management strategies. Overall, adopting LLMs throughout a user’s journey enables businesses to build deeper relationships with their customers than any competitor, leading to increased average time users spend with the business and more sales [59]. In e-commerce and retail, LLMs improve tailored recommendations and customer engagement by analyzing user preferences and behavioral data [30], [69], [76].

On the other hand, virtual shopping is one of the most widely used e-commerce applications, enabling users to interact with computer simulations to choose and buy virtual items. With LLM, users could interact with a computer shopping agent, using verbal communication or textual input, and then the agent receives the request, seeks clarification whether the request is vague, and successfully fulfills it. This process might involve retrieving and presenting information from a database, or, in the case of RSs, suggesting an item that best fits the shopper’s needs [90].

E. SOCIAL MEDIA & NETWORKS

Social media platforms derive substantial value from user interactions; therefore, personalized content recommendations become essential. The value of LLMs in this context is to predict users’ interests and preferences by analyzing their unstructured data, such as posts, comments, likes,



and shares [93]. Social media networks employ LLMs for sentiment analysis and the development of preference modeling [60], [82].

#### F. TOURISM & CULTURAL HERITAGE

LLMs can be trained to analyze vast amounts of textual data in museums, including exhibit descriptions, relevant literature, and external sources. Through this process, they can model the museum space as a whole in a unified way and, hopefully, serve visitors by providing more accurate, context-aware recommendations similar to those of experienced human guides. RS applications in the context of tourism and museums are relatively new and introduce unique challenges and opportunities [80]. This AI-powered tour guide, in turn, leads to increased visitor satisfaction and deeper engagement with the exhibits [94]. Therefore, in tourism and cultural heritage, LLMs facilitate collaborative cultural experiences and contextually informed advisory [95].

#### G. EDUCATION & LEARNING

The use of LLMs in building interactive and conversational systems has shown great potential, providing adaptive learning support and educational resources that benefit both teachers and students [96]. Additionally, integrating LLMs offers benefits, such as assisting students with course selection and academic planning [45]. These examples collectively demonstrate how LLMs are transforming various sectors by enabling more intelligent, context-aware, and user-centric computational systems.

#### H. MAGICAL

RSs have use-cases in different sectors, and one particularly interesting use-case is utilizing in museums, which has improved the level of enjoyment and satisfaction of visitors from these places because of the knowledge that can be transferred based on their preferences [95], [97]. LLM integration with RSs in this area can take available textual data in museums as input and can generate more accurate recommendations similar to an old-fashioned way of a human tour guide [80]. MAGICAL (Museum AI Guide for Augmenting Cultural Heritage with Intelligent Language model) is a digital tour guide that is AI-Powered, which has interaction with visitors [81]. This is based on the GPT4 model and have context of visitors behaviors and museum spaces, and also detailed exhibit information, which in results provides personalized and engaging recommendations [80].

### VII. CONCLUSION

This SLR examines the integration of LLMs into recommender systems, drawing on research from 2018 to 2025 to provide a comprehensive understanding of contemporary methodologies, applications, and emerging developments. The study focuses on three main research questions: identifying existing integration methods, developing a classification framework for LLM-enhanced recommender systems, and

highlighting the key domains in which recommender systems have been successfully integrated with LLMs.

The findings of this study indicate that LLM integration has significantly reshaped the landscape of recommender systems across multiple dimensions. The review identified five primary integration methodologies: (1) CLLM4Rec frameworks, which incorporate LLMs into traditional ID-paradigm systems; (2) knowledge-graph-enhanced approaches that improve interpretability and factual precision; (3) LLMERS architectures, which leverage the knowledge and model-enhancement capabilities of LLMs; (4) strategic placement of LLMs at various stages of the RS pipeline, including feature engineering, encoding, ranking, and user-system interaction; and (5) specialized frameworks such as LLM4Rec, which distinguish between discriminative and generative recommendation paradigms.

Moreover, the taxonomy analysis reveals that LLM-enhanced recommender systems can be classified along several key dimensions. Cold-start mitigation strategies allow systems to address user, item, and system cold-start challenges through approaches such as data augmentation and LLM-based simulation. The integration strategy dimension encompasses a range of implementation methods, including embedding-based and prompting-based integration, fine-tuned LLMs, and agentic or conversational systems. The system architecture dimension spans solutions from standalone LLM-driven recommenders to LLM-augmented hybrid systems and fully LLM-orchestrated platforms. Collectively, these classifications provide a structured foundation for understanding the diverse methodologies underpinning LLM-enhanced recommender systems.

The assessment-dimension analysis shows that traditional performance measures remain highly relevant, as LLM-based recommender systems achieve competitive accuracy while demonstrating superior ability to capture semantic and contextual nuances. Cold-start effectiveness improves substantially, with particularly large gains observed when leveraging textual information and retrieval-augmented generation techniques. Natural-language explanations further enhance explainability and interpretability, improving user experience and addressing critical concerns related to system transparency. However, efficiency remains a major challenge due to the substantial computational demands of LLMs, underscoring the need for techniques such as model distillation and parameter-efficient fine-tuning to improve practicality and scalability.

Additionally, this study examines the current utilization of LLMs in recommender systems across a range of industries. LLMs have emerged as a transformative technology, enabling organizations to navigate increasingly complex decision-making processes while meeting customer expectations for highly personalized and interactive experiences.

Recent studies show that LLM-based systems are transforming multiple application domains. In e-commerce and retail, they optimize product recommendations, personalize marketing strategies, and enhance customer engagement.

**TABLE 6.** Primary studies in LLM-enhanced recommender systems.

| S# | Article Reference | Focus Area / Contribution                                                          |
|----|-------------------|------------------------------------------------------------------------------------|
| 1  | [5]               | Interactive and explainable LLM-augmented recommender systems                      |
| 2  | [11]              | LLMs as zero-shot rankers for recommendations                                      |
| 3  | [69]              | Recommendation as instruction following with LLMs                                  |
| 4  | [60]              | LLMs as zero-shot conversational recommenders                                      |
| 5  | [12]              | Generative pretrained language models for open-ended recommendations               |
| 6  | [51]              | Parameter-efficient prompt tuning for scale                                        |
| 7  | [76]              | Collaborative LLM for recommender systems                                          |
| 8  | [21]              | Impact of LLMs on recommender systems: extensive review                            |
| 9  | [7]               | LLM-enhanced recommender systems survey                                            |
| 10 | [22]              | How recommender systems benefit from LLMs survey                                   |
| 11 | [17]              | Unifying LLMs and knowledge graphs roadmap                                         |
| 12 | [50]              | Graph-of-thought using LLMs for complex problems                                   |
| 13 | [24]              | Hallucination in natural language generation survey                                |
| 14 | [55]              | ChatGPT multitask evaluation on reasoning and hallucination                        |
| 15 | [8]               | LLM4Rec: Comprehensive survey on LLM integration in RS                             |
| 16 | [26]              | Explainability for large language models survey                                    |
| 17 | [62]              | Small language models for data-centric cold-start recommendation                   |
| 18 | [63]              | LLMs for user interests interpretation in cold-start items                         |
| 19 | [13]              | LLM simulator for cold-start recommendation                                        |
| 20 | [16]              | LLMTreeRec: Unleashing LLM power for cold-start recommendations                    |
| 21 | [65]              | LLMs as data augmenters for cold-start item recommendation                         |
| 22 | [71]              | Prompt engineering potential for LLMs                                              |
| 23 | [70]              | Robust similar legal case recommender system with LLM                              |
| 24 | [72]              | Reinforced prompt personalization for recommendation                               |
| 25 | [75]              | Large language and graph assistant (LLAGA)                                         |
| 26 | [68]              | Recommender AI agent: Integrating LLMs for interactive recommendations             |
| 27 | [19]              | Recommender systems in the era of LLMs                                             |
| 28 | [64]              | Novel framework leveraging LLMs for cold-start advertising                         |
| 29 | [80]              | LLMs as recommender systems in museums                                             |
| 30 | [81]              | Crafting museum guides using ChatGPT-4                                             |
| 31 | [83]              | Fine-tuned LLMs for reducing data needs in load profile analysis                   |
| 32 | [18]              | Agentic recommender systems in era of multimodal LLMs                              |
| 33 | [67]              | Large language model enhanced conversational recommender system                    |
| 34 | [20]              | Comprehensive survey on LLM-powered recommender systems                            |
| 35 | [73]              | Recranker: Instruction tuning LLM as ranker for top-k recommendation               |
| 36 | [74]              | Enhanced recommendations with retrieval-augmented LLM                              |
| 37 | [10]              | LLM-enhanced recommender systems: Methods, applications and trends                 |
| 38 | [85]              | Learning human feedback from LLMs for quality-aware recommendation                 |
| 39 | [82]              | Harnessing LLMs for dynamic intention understanding in recommenders                |
| 40 | [77]              | One model for all: LLMs as domain-agnostic recommender systems                     |
| 41 | [15]              | DALLRec: Data augmentation framework with fine-tuned LLM                           |
| 42 | [6]               | Survey of large language models                                                    |
| 43 | [34]              | Unified multi-task learning for multi-goal conversational recommenders             |
| 44 | [33]              | Generative recommendation: Next-generation recommender paradigm                    |
| 45 | [35]              | Towards unified conversational recommenders via knowledge-enhanced prompt learning |
| 46 | [36]              | RecInDial: Unified framework for conversational recommendation                     |
| 47 | [37]              | Topic-guided conversational recommender system                                     |
| 48 | [84]              | Breaking barrier: LLMs for industrial recommendations via knowledge graphs         |
| 49 | [98]              | Exploring limits of transfer learning with unified text-to-text transformer        |
| 50 | [46]              | Survey of transformers and ensemble learning for NLP                               |
| 51 | [53]              | ClinicalBERT: Modeling clinical notes and predicting readmission                   |
| 52 | [54]              | Infusing behavior science into LLMs for activity coaching                          |
| 53 | [59]              | Application of LLMs in recommender systems                                         |
| 54 | [45]              | LLM-based literature recommender system in higher education                        |
| 55 | [66]              | Advances and challenges in conversational recommender systems                      |

In news and media platforms, LLMs support summarization, fact extraction, and tailored content selection. LLMs have also seen extensive use in the healthcare domain, where they assist clinicians by analyzing medical records, supporting diagnosis, and improving patient-provider interactions. Additional domains—such as social media, tourism,

and cultural heritage—employ LLMs to analyze user behavior, detect sentiment, customize content streams, and deliver context-aware guidance and interactive visitor experiences.

The growing adoption of LLMs demonstrates their ability to address long-standing challenges in recommender systems,

including cold-start problems, sparse interaction histories, and evolving user preferences. In summary, LLMs are reshaping the landscape of personalized recommendations in a profound way.

### VIII. EMERGING SUBFIELDS AND FUTURE WORK

The rapid integration of LLMs in recommender systems has led to the emergence of many subfields. These new directions indicate a transition from static data-driven recommendation models to more adaptive, interactive, and intelligent frameworks. These emerging subfields are outlined in Section II, part C, LLMs in enhancing recommender systems. They can also be summarized in the following thrusts: Agentic Recommender Systems, Multimodal Recommender Systems, Retrieval-Augmented Recommender Systems, Human-in-the-Loop and Feedback-Driven Recommender Systems, Universal and Cross-Domain Recommender Systems, and Explainable and Trustworthy Recommender Systems.

Although significant advancements were identified in the studies under review, numerous limitations remain in current research on LLM-enhanced recommender systems. First, most existing studies rely on small-scale or highly domain-specific datasets, which limits the generalizability of findings to broader, real-world environments. Many evaluations are conducted in controlled experimental settings rather than active, dynamic user scenarios, reducing ecological validity. Second, the computational cost of LLM inference presents a substantial barrier, making large-scale or real-time recommendation pipelines challenging for many organizations. Third, the field lacks standardized evaluation practices; studies vary widely in datasets, metrics, and baseline models, making direct comparison and benchmarking difficult.

Furthermore, although explainability has received notable attention, there remains a significant lack of systematic investigation into fairness, bias propagation, and ethical considerations. Similarly, there is a clear need to develop standardized user-centered evaluation approaches, such as satisfaction surveys, perceived recommendation quality, task completion rates, and controlled A/B testing. These measures are important for closing the gap between strong offline performance and actual user satisfaction in real-world settings. Finally, integration of structured knowledge sources, dynamic user modeling, and cross-domain transfer capabilities remains underexplored, limiting system adaptability and reducing effectiveness in supporting long-term personalization.

Future research must address these limitations by developing methodologies that are both more scalable and contextually aware. Establishing standardized benchmark datasets and evaluation protocols is essential to ensure fair comparison, reproducibility, and meaningful progress across studies. Improving computational efficiency through techniques such as model distillation, quantization, mixture-of-experts architectures, and hardware acceleration will be critical for enabling practical, real-time deployment.

Research should also explore privacy-preserving designs, including federated learning and differential privacy, to safeguard user data while still leveraging the capabilities of LLMs. Additionally, future studies need to systematically examine the influence of LLM-induced biases on recommendation outcomes and develop robust mitigation strategies to improve fairness and ethical reliability.

Enhanced integration of multimodal signals (e.g., text, images, audio), knowledge graphs, and temporal preference modeling can support deeper personalization and richer contextual understanding. Finally, advancements in cross-domain transfer, few-shot learning, and long-tail item recommendation will improve the robustness and adaptability of LLM-enhanced recommender systems, aligning them more closely with real-world complexity and evolving user expectations.

### APPENDIX PRIMARY STUDIES IN LLM-ENHANCED RECOMMENDER SYSTEMS

See Table 6.

### REFERENCES

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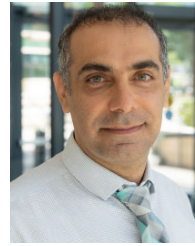
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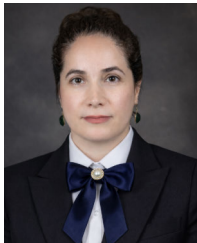
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